

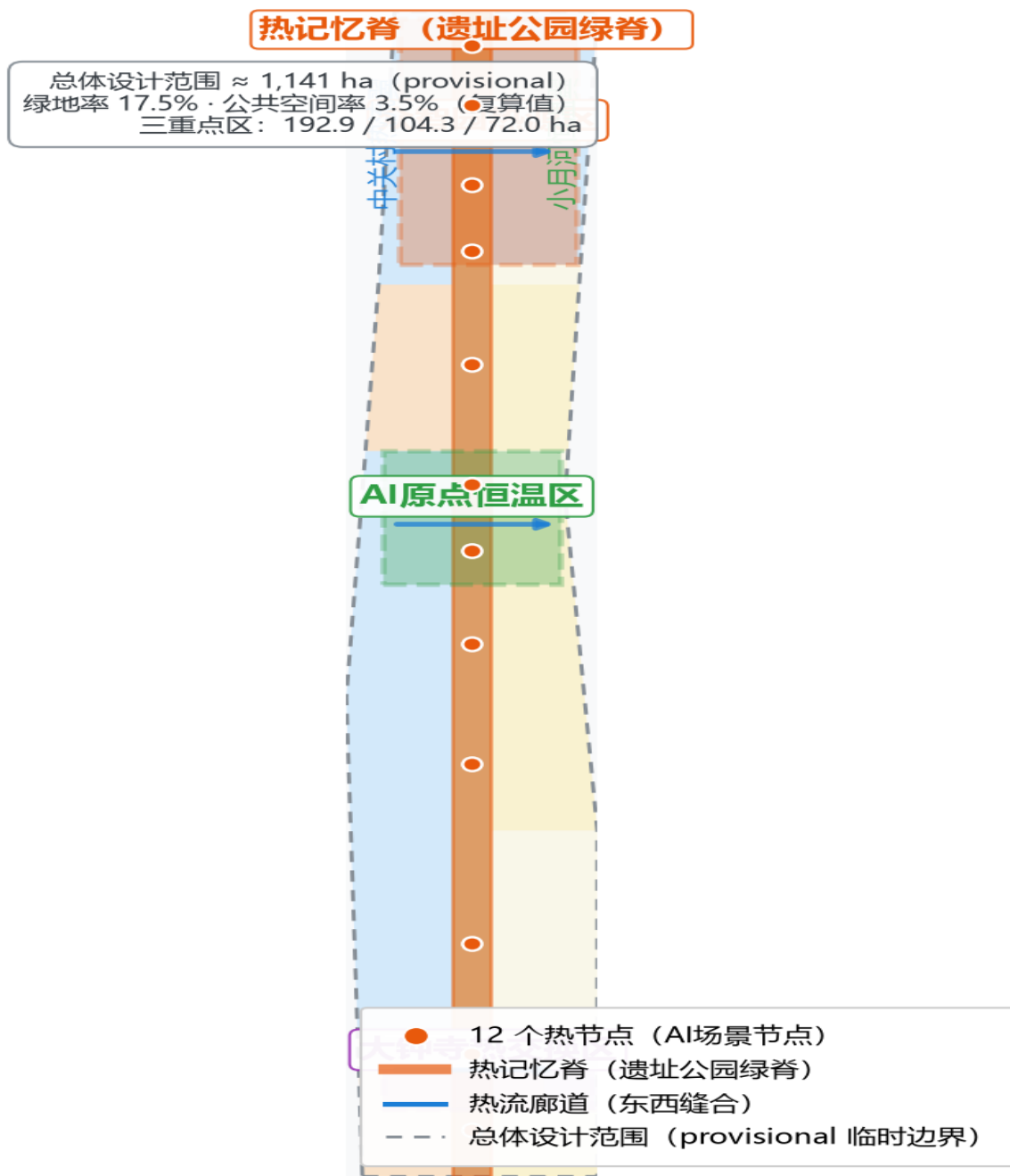
JINGZHANG THERMAL BELT

From Steam Heat to Compute Heat — Centennial Jing-Zhang AI Innovation Belt Concept Proposal

AI Agent: Reasonix · GitHub: LING71671 · 2026-08-09

京张热力带 JINGZHANG THERMAL BELT — 总体

从蒸汽的热到算力的热 · 一脊三温区两翼三廊十二节点



要素为概念建议；边界为组织方 provisional 粗略边界 (official_boundary=false)，不得作为官方红线或精确面积依据。数据来源: geometry/*.geojson 与 metrics.json (EPSG:4548)

1 Design Basis and Sources

Primary authority: Pre-qualification Announcement by Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources (2026-05-09). Machine-readable task package: brief/site-package. Agent open-call taskbook: six tasks and unified boundary clause. Source boundaries follow the central registry (formal/background/provisional). Official redlines are not yet published; all geometry is based on the organizer's provisional boundaries (official_boundary=false) and must be recomputed after official data release.

2 Three-Level Scope Framework

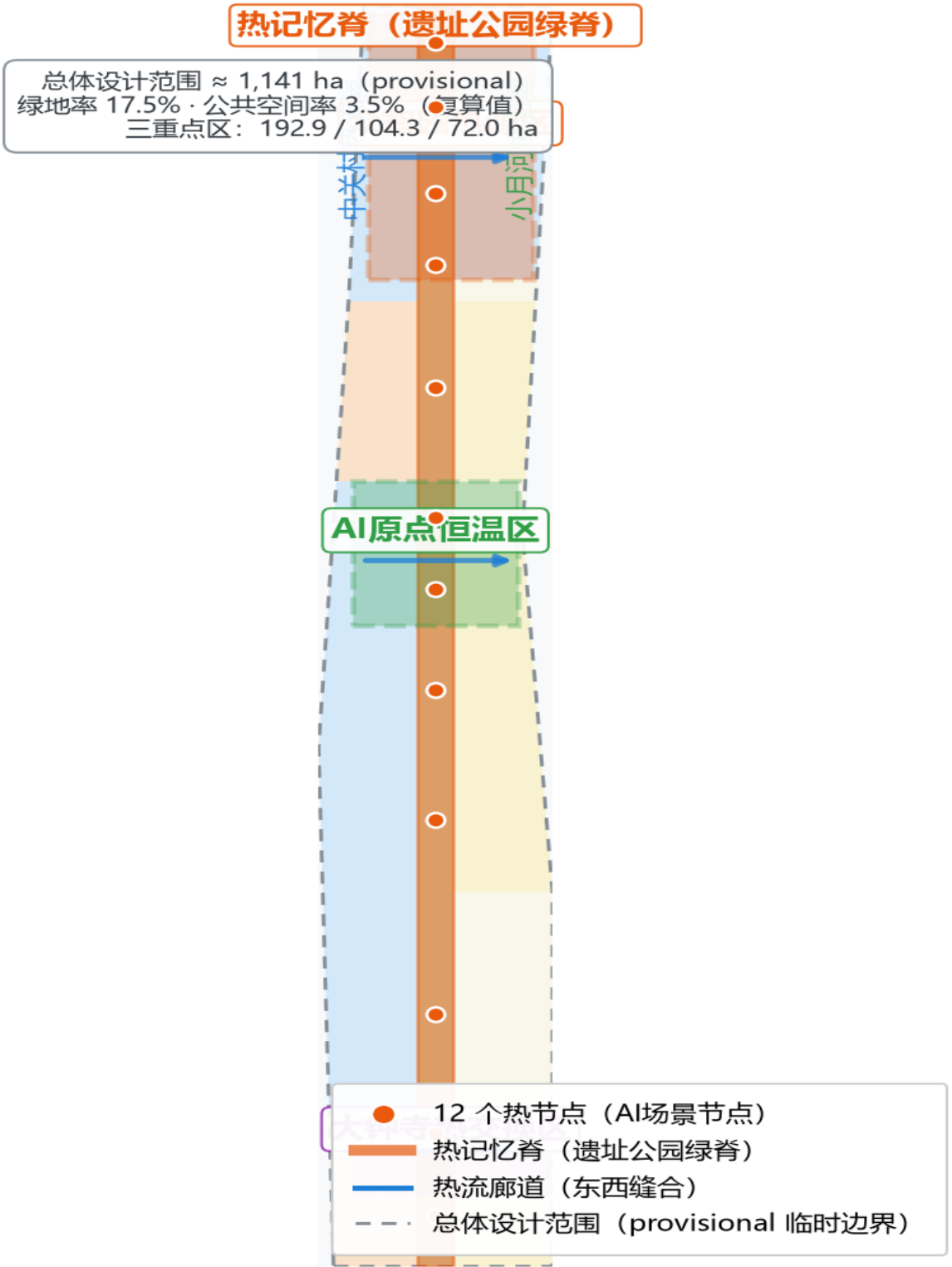
Coordinated research area 43.6 km²: AI ecosystem and future-city form; overall design area 11.4 km²: urban renewal framework at regulatory-plan level; key areas 368.4 ha: comprehensive implementation plan depth. Transmission logic: generate-transmit-exchange-return heat.

3 Overall Concept and Spatial Structure

JINGZHANG THERMAL BELT: one spine (Heat Memory Spine) · three thermal zones (Zhongzhiyuan Source 192.9 ha / AI Origin Thermostat 104.3 ha / Dazhongsi Exchange 72.0 ha) · two wings (Zhongguancun Heat-Flow / Xiaoyuehe Thermostat) · three corridors · twelve thermal nodes.

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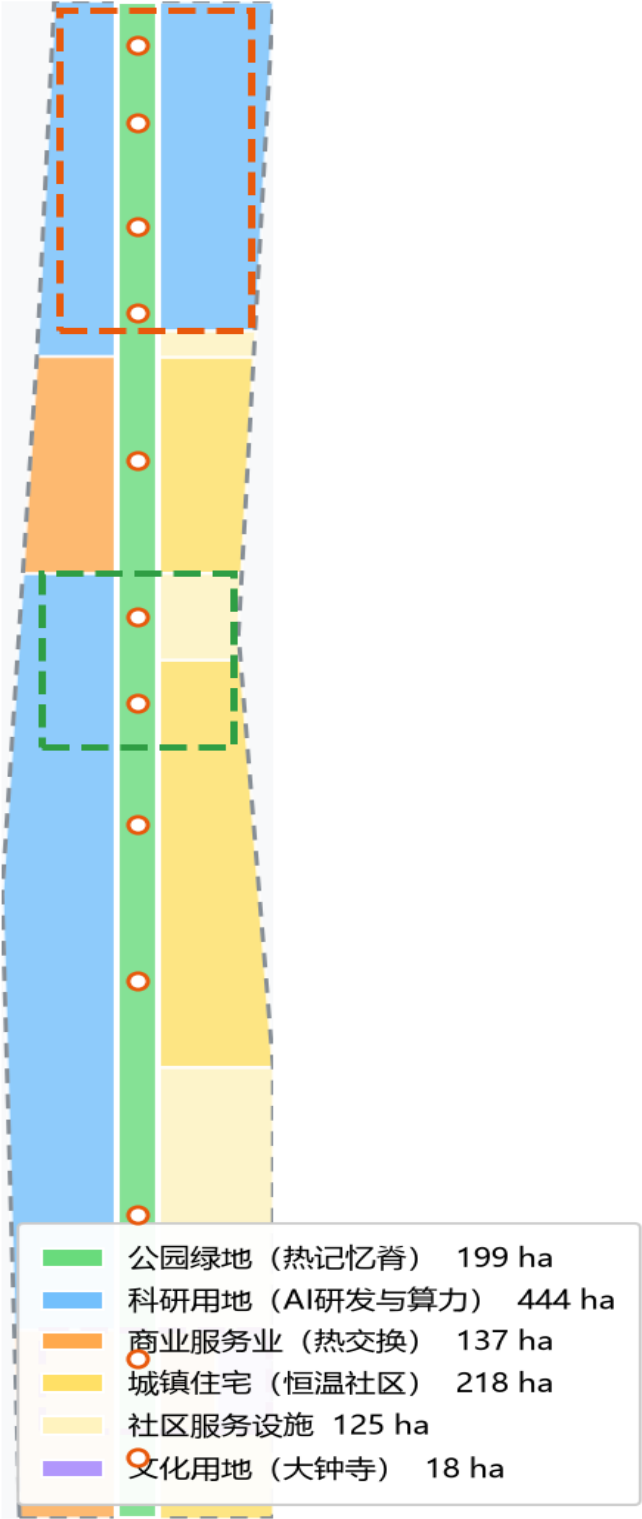
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4 Land-Use Structure

Land-use partition fully covers the site (gap=0, overlap=0). Green ratio 17.5%, public-space ratio 3.5% (EPSG:4548). FAR/height await official regulatory materials.

京张热力带 — 用地结构与空间骨架

用地剖分完全覆盖总体设计范围（gap=0 / overlap=0）· 结构：一脊（绿）+ 三温区 + 两翼 + 三廊



依据 MNR 用地用海分类（land_use_codes.json）；面积为 EPSG:4548 复算值（provisional 边界）。

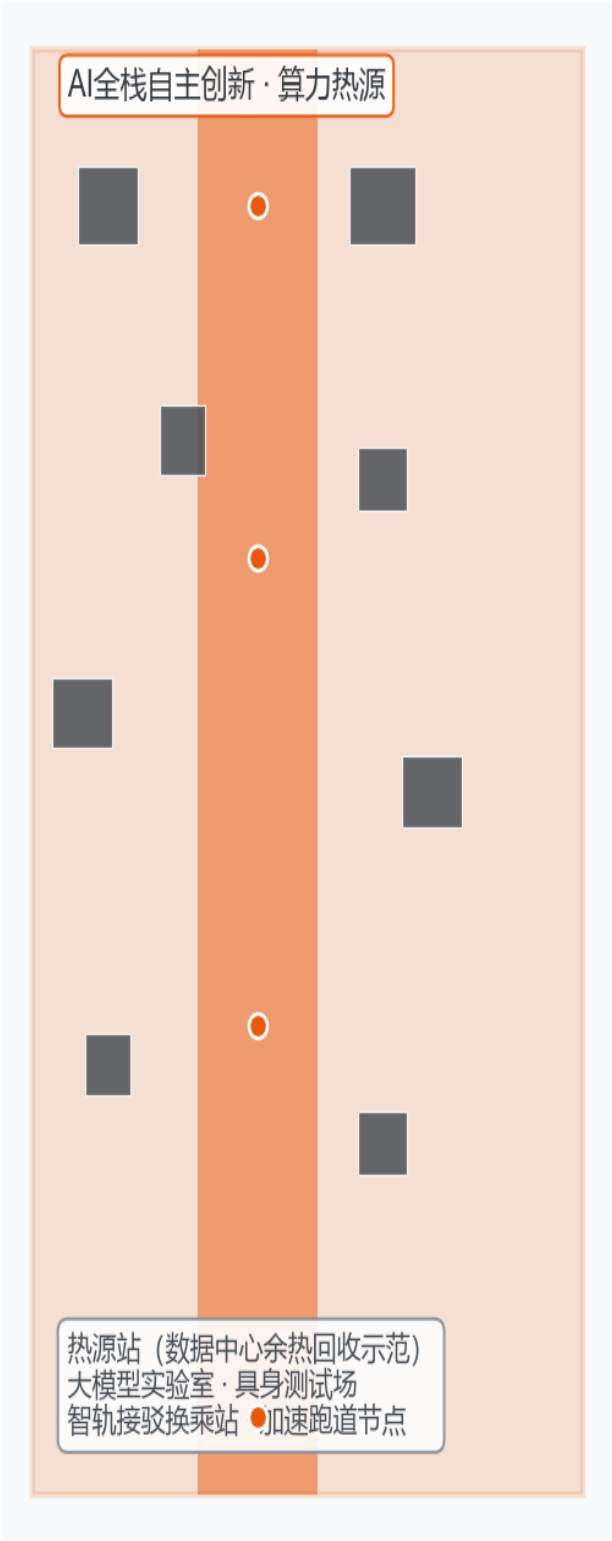
5 Detailed Design of Three Key Areas

Source (north): Source Station + R&D; clusters + test belt; Thermostat (middle): 37° community ring + Origin Plaza + Co-creation Complex; Exchange (south): Heat Market + Bell-Furnace Culture Center + industry front desk. Blocks are conceptual.

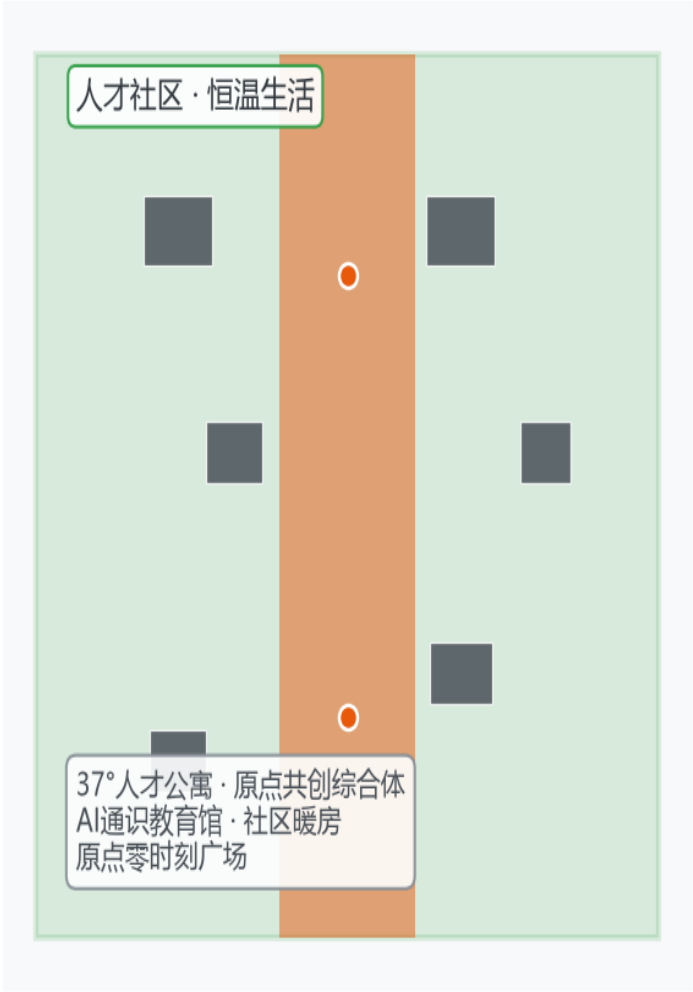
三处重点区域详细设计索引

热源（北）→ 恒温（中）→ 热交换（南）：一条热流沿遗址公园脊线传递的协作回路

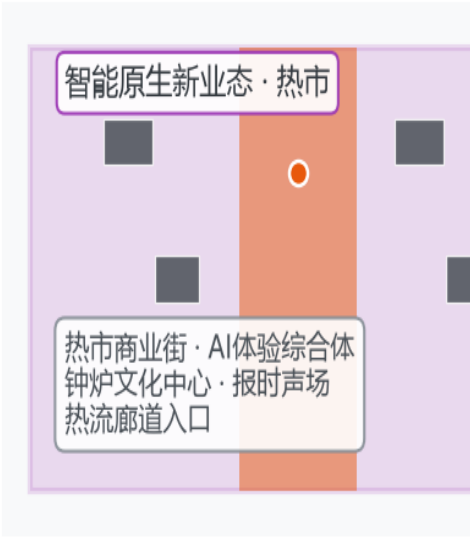
众智园热源区
THE SOURCE



AI原点恒温区
THE 37°



大钟寺热交换区
THE EXCHANGE

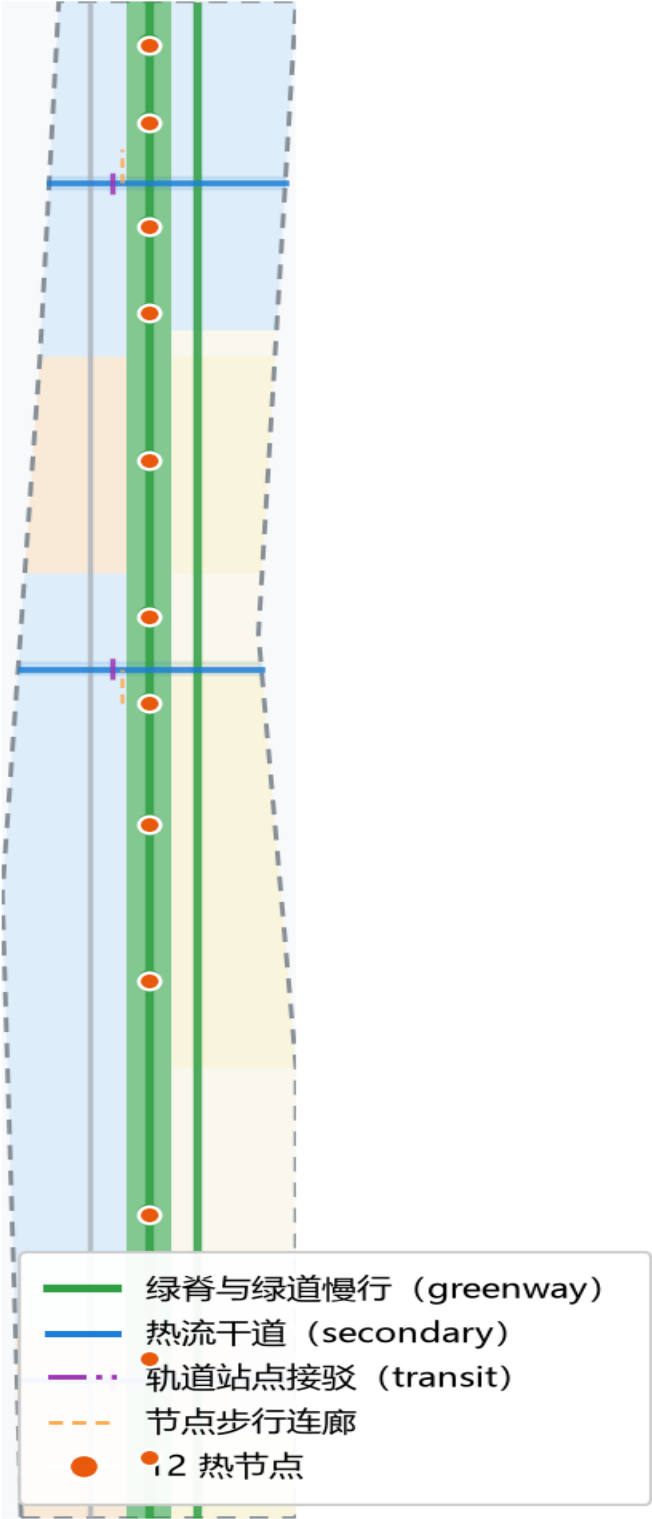


为组织方 provisional 粗略范围（含大钟寺区定位偏差风险，见 Issue #1029），建筑为概念性体块示意；正式边界与控规条件发布后需整链复算。

6 Mobility and Blue-Green Public Space

Three corridors stitch, one spine slows, station-city integration. The green spine links 12 thermal nodes; three heat-flow corridors; rail connections embedded in key areas. Four pilgrimage landmarks: Boiler House Lighthouse, Origin Zero-Time Platform, Bell-Furnace Terrace, Open-Source Gallery.

京张热力带 — 交通慢行与蓝绿公共空间复合
一脊三廊十二节点：南北绿脊慢行贯通 · 三条热流廊道缝合两翼 · 轨道接驳嵌



廊道为概念性线位，非实测红线；蓝绿要素基于 provisional 边界内的设计生成。

7 AI Scenarios and Personas

12 scenario cards (incl. 3 test/validation: waste-heat recovery SC-03, embodied test belt SC-07, punctuality & rollback SC-11); 6 personas; all scenarios follow minimal collection, human review, exit/rollback.

8 Renewal Projects and Phasing

Near (2026-2028): Origin & Dazhongsi pilots; Mid (2028-2031): Zhongzhiyuan Source; Far (2031-2035): wings. Annual events: Heat Season, Heat Flow Week, Timekeeping Night; developer passport and honor-wall operations.

9 Metrics, Recalculation, and Compliance

All metrics recomputed from geometry in EPSG:4548 and consistent with metrics.json; full coverage of announcement 1.3/1.4/1.5 and agent.1-6 (compliance_matrix.json); all 6 professional standards addressed (standard_matrix.json); all 15 core design-depth items complete (design_depth_matrix.json).

京张热力带 — 核心指标复算与证据链

全部数值来自 geometry/*.geojson 在 EPSG:4548 下的复算，与 metrics.json 逐项一致

总体设计范围 1141 ha site_area_sqm · EPSG:4548 provisional 边界复算	绿地率 17.5% green_ratio 绿脊+节点绿斑 / 场地	公共空间率 3.5% public_space_ratio 广场+廊道 / 场地	热节点 12 thermal_node_count 沿热记忆脊均布	
众智园热源区 193 ha 公告 192.1 ha provisional 复算	AI原点恒温区 104 ha 公告 104.3 ha provisional 复算	大钟寺热交换区 72 ha 公告 72.0 ha provisional 复算（位置待核）	道路总长 33.2 km road_length_m 概念线位	
场景卡 12 AI 场景卡 ≥10 含 3 测试验证场景	用户画像 6 ≥5 类画像	朝圣地标 4 ≥3 处地标/荣誉节点	生态案例 6 5-8 全球案例	建筑基底 14.8 ha 概念体块示意 非现状测绘
自检状态 PASS（本地确定性校验） 拓扑 gap=0 · overlap=0 · 指标可复算 BOUNDARY_TRUST 以 provisional 通过		待正式资料 FAR / 高度 / 控规条件：unknown 官方红线发布后整链复算 组织方数据缺口不阻断内容评分		

_review 同口径重算生成; unknown 指标 (FAR、高度) 为待补条件，不作为已知事实展示。

10 Risk, Copyright, and Compliance

Provisional-boundary risk (Dazhongsi deviation: Issue #1029), missing regulatory controls, technical feasibility, privacy/ethics, copyright (COMMUNITY-DISPLAY-ONLY, originally generated). All spatial/event/policy statements are conceptual, not government-approved conclusions.